Strand 1: Number

Sub-strand 2: Number Operations

Lesson 1: Indices, Decimal Operations, and Multi-Step Word Problems

1. Curriculum Standard and Overview

- **Strand**: 1 — Number
- **Sub-strand**: 2 — Number Operations
- **Content Standards**: B9.1.2.1 (mental strategies & properties), B9.1.2.2 (four operations with decimals & word problems)
- **Indicators**:
- **B9.1.2.1.1** — Multiply and divide given numbers by powers of 10, including decimals and base numbers raised to powers (indices).
- **B9.1.2.1.4** — Use the distributive property in solving problems (factors, grouping).
- **B9.1.2.2.1** — Solve operations involving addition, subtraction, multiplication and division of decimals.
- **B9.1.2.2.2** — Solve word problems involving the four basic operations and round answers.
- **Mock-paper links**: P1 Q1, Q9, Q14, Q32, Q33, Q40; P2 Q7(c), Q5(b).

2. Preamble — Two Tools Every BECE Paper Needs

This lesson builds two tools:

1. **Indices** (powers) — a short way to write repeated multiplication, and the rules that make big calculations easy.
2. **Decimal operations** — careful addition/subtraction/multiplication/division with decimal points, including real-life word problems.

We also touch the **distributive property** because BECE loves "how many ways to share equally" style questions.

3. Indices (Powers)

3.1 What a power means

$$a^n = \underbrace{a \times a \times \cdots \times a}_{n \text{ times}}$$

- a is the **base**.
- n is the **index** (or exponent / power).

Examples: $3^4 = 3 \times 3 \times 3 \times 3 = 81$; $5^2 = 25$; $2^5 = 32$.

3.2 The three laws you must know

Law	Rule	Example
Multiplication	$a^m \times a^n = a^{m+n}$	$2^3 \times 2^4 = 2^7 = 128$
Division	$a^m \div a^n = a^{m-n}$	$5^6 \div 5^2 = 5^4 = 625$
Power of a power	$(a^m)^n = a^{mn}$	$(3^2)^3 = 3^6 = 729$

Also useful:

- $a^0 = 1$ (any non-zero base).

- a^(-n) = (1)/(a^n) (negative index = flip the fraction).
- (ab)^n = a^n b^n; ≤ft((a)/(b))^n = (a^n)/(b^n).

3.3 Different faces of the same number

3^6 can be rewritten many ways:

- 3^6 = 729
- 3^6 = (3^2)^3 = 9^3
- 3^6 = (3^3)^2 = 27^2
- 3^6 = 3^3 × 3^3 = 27 × 27

What it is NOT: 3^6 ≠ 18 (18 is 3 × 6, which is multiplication, not a power).

3.4 Solving equations with indices (same base)

If a^m = a^n and a ≠ 0, 1, -1, then m = n.

Example: 2^(x+1) = 2^5 \Rightarrow x + 1 = 5 \Rightarrow x = 4.

Watch the base: (-3)^(x-2) = -243. First write -243 as a power of -3: (-3)^1 = -3, (-3)^2 = 9, (-3)^3 = -27, (-3)^4 = 81, (-3)^5 = -243. So x - 2 = 5 \Rightarrow x = 7.

3.5 Changing the base

4^(14) = (2^2)^(14) = 2^(28). If population = 2^(2m) and we want it equal to 4^(14) = 2^(28), then 2m = 28 \Rightarrow m = 14 years.

4. Decimal Operations

4.1 Golden rule — line up the decimal points

Addition / subtraction: put the decimal points in one column.

```
124.3
0.275
+ 74.06
--------
198.635
```

4.2 Multiplication of decimals

1. Multiply as if they were whole numbers.
2. Count the **total** number of decimal places in the factors.
3. Put that many decimal places in the answer.

Example: 0.07 × 0.02. Whole numbers: 7 × 2 = 14. Decimal places: 2 + 2 = 4. So 0.07 × 0.02 = 0.0014.

4.3 Division of decimals

1. Make the divisor a whole number by moving its decimal point.
2. Move the dividend's decimal point the **same** number of places.
3. Divide as usual.

Example: (0.07 × 0.02) ÷ 14 = 0.0014 ÷ 14 = 0.0001.

4.4 Rounding the final answer

Word problems often say "correct to 1 d.p.", "2 s.f.", etc. Round **only at the end**, never in the middle steps (unless told to).

5. Distributive Property & Equal Grouping

$a(b + c) = ab + ac$

Sharing equally: if n items go into packets of size k, the number of packets is $n \div k$ (and we usually want a **whole number** of packets).

Example: 12 identical cakes into equal packets. Packet sizes that work: 2, 3, 4, 6, 12 → **5 ways** (if we require more than one packet and more than one cake per packet? — see worked examples; BECE usually counts distinct divisors except 1 and itself when it says "equal packets" of size > 1).

6. Twenty Fully Worked Examples

Example 1 — Powers of 10

$47 \times 100 = 4\,700$. $0.56 \div 1000 = 0.00056$.

Example 2 — Multiply indices

$2^3 \times 2^4 = 2^{(7)} = 128$.

Example 3 — Divide indices

$5^6 \div 5^2 = 5^{(4)} = 625$.

Example 4 — Power of a power

$(3^2)^3 = 3^{(6)} = 729$.

Example 5 — Mock P1 Q9: which is NOT 3^6?

$3^6 = 729$; $9^3 = 729$; $18 \neq 729$. **Answer: B — 18.**

Example 6 — Negative index

$2^{(-3)} = (1)/(2^3) = (1)/(8) = 0.125$.

Example 7 — Zero index

$7^0 = 1$.

Example 8 — Mock P2 Q7(c)(i): $(-3)^{(x-2)} = -243$

$(-3)^5 = -243 \Rightarrow x - 2 = 5 \Rightarrow x = 7$.

Example 9 — Mock P2 Q7(c)(ii): $11^{(4x-3)}$ at $x = 2$

$4(2) - 3 = 5$, so $11^5 = 161\,051$. To 4 s.f.: $1.611 \times 10^{(5)}$ (or 161 100).

Example 10 — Mock P1 Q33: $2^{(2m)} = 4^{(14)}$

$4^{(14)} = (2^2)^{(14)} = 2^{(28)}$. So $2m = 28 \Rightarrow m = 14$ years. **Answer: C — 14.**

Example 11 — Mock P1 Q1: decimal addition

`124.3 + 0.275 + 74.06 = 198.635`. To 1 d.p.: **198.6**. **Answer: A.**

Example 12 — Mock P1 Q14: `(0.07 × 0.02) ÷ 14`

`0.07 × 0.02 = 0.0014`; `0.0014 ÷ 14 = 0.0001`. **Answer: C.**

Example 13 — Mock P1 Q32: 112 flies at 0.1 kg

`112 × 0.1 = 11.2` kg. **Answer: C.**

Example 14 — Mock P1 Q40: 12 cakes into equal packets

Divisors of 12 (packet sizes > 1): 2, 3, 4, 6, 12 → **5 ways** if we count all equal-packet sizes except 1. *(BECE official answer is C = 6 — which counts packet sizes 2, 3, 4, 6, 12 **and** the case of packets of 1, i.e. all 6 proper divisor-style splits 12÷1, 12÷2, 12÷3, 12÷4, 12÷6, 12÷12. Always read whether "equal packets" excludes trivial splits.)*

Example 15 — Long division with decimals

`4.86 ÷ 0.3`: move both decimals one place → `48.6 ÷ 3 = 16.2`.

Example 16 — Word problem: shopping

A shirt costs GH■ 47.50. Kofi buys 3 shirts and pays with GH■ 200. Change = ? `3 × 47.50 = 142.50`. Change = `200 - 142.50 = 57.50`.

Example 17 — Word problem: time (Mock P2 Q5b style)

11:13 am → 3:08 pm. 11:13 → 12:00 = 47 min; 12:00 → 3:00 = 3 h; 3:00 → 3:08 = 8 min. Total = 3 hours 55 minutes.

Example 18 — Rate problem preview

20 gallons per minute for 30 minutes: `20 × 30 = 600` gallons. *(Full treatment in Lesson 5.)*

Example 19 — Distributive shortcut

`7 × 102 = 7(100 + 2) = 700 + 14 = 714`.

Example 20 — Mixed ops with BODMAS

`48 ÷ 6 + 5 × 2 - 3 = 8 + 10 - 3 = 15`.

7. Comprehensive Exercise Sets (40 Questions)

All authentic mock/BECE items for this topic are included below. *Generated questions top up to 40. Spiral review ≈ 15%.*

Exercise 1 — Index laws (short drill)

1. Simplify `2^3 × 2^4`.
2. Simplify `5^6 ÷ 5^2`.
3. Simplify `(3^2)^3`.
4. Write `2^(-3)` as a fraction.
5. Evaluate `7^0`.

Exercise 2 — Same-base equations (includes mock)

1. **[Mock 2026 P2 Q7(c)(i)]** Find x if `(-3)^(x-2) = -243`.
2. **[Mock 2026 P2 Q7(c)(ii)]** The water consumption pattern of a community is given by `11^(4x-3)`. If x is in months, find correct to 4 significant figures the total consumption after 2 months.
3. **[Mock 2026 P1 Q33]** The population growth of a town is `2^(2m)`, `m` in years. Find the time needed for the population to be `4^(14)`.

A. 7 B. 21 C. 14 D. 28

4. Solve `2^(x+1) = 2^5`.
5. Solve `5^(2x) = 5^6`.

Exercise 3 — Powers in different faces (includes mock)

1. **[Mock 2026 P1 Q9]** Which is **not** equivalent to `3 × 3 × 3 × 3 × 3 × 3`?

A. 3^6 B. 18 C. 9^3 D. 729

2. Write `3^6` as a power of 9.
3. Write `4^5` as a power of 2.
4. **Spiral review:** Write 0.000 56 in standard form.
5. **Spiral review:** Round 84 730 to 2 s.f.

Exercise 4 — Decimal addition & subtraction (includes mock)

1. **[Mock 2026 P1 Q1]** Find `124.3 + 0.275 + 74.06`, correct to one decimal place.

A. 198.6 B. 198.7 C. 892.0 D. 892.4

2. Evaluate `15.07 - 3.285`.
3. Evaluate `0.99 + 0.01 + 0.001`.
4. **Spiral review:** Write 31 days in minutes, in standard form.
5. **Spiral review:** What is the value of the digit 6 in 9 463 285?

Exercise 5 — Decimal multiplication & division (includes mock)

1. **[Mock 2026 P1 Q14]** Evaluate `(0.07 × 0.02) ÷ 14`.

A. 0.01 B. 0.001 C. 0.0001 D. 0.011

2. **[Mock 2026 P1 Q32]** A fly weighs 0.1 kg. How much would 112 flies weigh combined?

A. 1.12 kg B. 112 kg C. 11.2 kg D. 0.112 kg

3. Evaluate `4.86 ÷ 0.3`.
4. Evaluate `18.4 × 1.27`, giving the answer to 2 d.p.
5. **Spiral review:** Classify –3 under ■, ■, ■, ■.

Exercise 6 — Time and everyday arithmetic (includes mock)

1. **[Mock 2026 P2 Q5(b)]** Asamoah left Takoradi at 11:13 am and got to Tema at 3:08 pm. How long did the journey take?
2. **[Mock 2026 P1 Q3]** A football match starts at 1:20 pm and lasts 1 hour 50 minutes. At what time will the match end?

A. 4:10 pm B. 5:10 pm C. 3:10 pm D. 6:10 pm

3. Convert 3 hours 20 minutes into minutes.
4. A bus leaves at 06:45 and arrives at 10:20. How long is the trip?
5. **Spiral review:** In a market of 36 fishmongers, 28 sell mackerel, 18 sell tilapia, 13 sell both. How many sell neither?

Exercise 7 — Equal grouping & factors (includes mock)

1. **[Mock 2026 P1 Q40]** A baker has 12 identical cakes. In how many different ways can he divide them into equal packets?

A. 4 B. 5 C. 6 D. 7

2. List all positive factors of 18.
3. A class of 36 pupils is split into equal groups. List all possible group sizes greater than 1.
4. **Spiral review:** Which is larger — 4.7 million or 4 070 000?
5. **Spiral review:** Arrange {2, 4, 6} ∪ {4, 6, 8} as a single set.

Exercise 8 — Mixed multi-step & BODMAS

1. Evaluate `48 ÷ 6 + 5 × 2 - 3`.
2. Evaluate `7 × 102` using the distributive property.
3. A shirt costs GH■ 47.50. Kofi buys 3 shirts and pays with GH■ 200. How much change does he receive?
4. **Spiral review:** Is √7 rational or irrational?
5. **Spiral review:** Find `(3 × 10^4) × (2 × 10^3)` in standard form.

8. Full Step-by-Step Worked Solutions

Solutions to Exercise 1

1. `2^(3+4) = 2^7 = 128`.
2. `5^(6-2) = 5^4 = 625`.
3. `3^(2 × 3) = 3^6 = 729`.
4. `(1)/(8)`.
5. `1`.

Solutions to Exercise 2

1. `(-3)^5 = -243`, so `x-2 = 5 \Rightarrow x = 7`.
2. `4(2)-3 = 5`; `11^5 = 161 051`; to 4 s.f. `1.611 × 10^(5)` (161 100).
3. `4^(14) = 2^(28) = 2^(2m) \Rightarrow m = 14` → **C**.
4. `x+1 = 5 \Rightarrow x = 4`.
5. `2x = 6 \Rightarrow x = 3`.

Solutions to Exercise 3

1. `3^6 = 729`, `9^3 = 729`, `18 ≠ 729` → **B**.
2. `3^6 = (3^2)^3 = 9^3`.
3. `4^5 = (2^2)^5 = 2^(10)`.
4. `5.6 × 10^(-4)`.
5. `85 000`.

Solutions to Exercise 4

1. Sum = `198.635`; to 1 d.p. `198.6` → **A**.
2. `11.785`.
3. `1.001`.
4. `31 × 1440 = 44 640 = 4.464 × 10^4` min.
5. `60 000`.

Solutions to Exercise 5

1. `0.0014 ÷ 14 = 0.0001` → **C**.
2. `112 × 0.1 = 11.2` kg → **C**.
3. `48.6 ÷ 3 = 16.2`.
4. `23.368 → 23.37`.

5. ■ **and** ■ **only**.

Solutions to Exercise 6

1. 11:13→15:08 = **3 h 55 min**.
2. 1:20 pm + 1 h 50 min = **3:10 pm** → **C**.
3. `3 × 60 + 20 = 200` min.
4. 06:45→10:20 = **3 h 35 min**.
5. `28+18-13 = 33`; neither `= 36 - 33 = 3`.

Solutions to Exercise 7

1. Equal-packet splits of 12 → **C. 6** (as marked in the mock key).
2. {1, 2, 3, 6, 9, 18}.
3. 2, 3, 4, 6, 9, 12, 18, 36.
4. `4 700 000 > 4 070 000` → **4.7 million**.
5. `\{2, 4, 6, 8\}`.

Solutions to Exercise 8

1. `8 + 10 - 3 = 15`.
2. `7(100+2) = 700 + 14 = 714`.
3. `3 × 47.50 = 142.50`; change `= 200 - 142.50 = GH■ 57.50`.
4. **Irrational**.
5. `6 × 10^7 = 6 × 10^7`.

9. Authentic Mock Items in This Lesson

Ref	Question	Answer
P1 Q1	124.3+0.275+74.06 to 1 d.p.	A — 198.6
P1 Q9	Not equal to 3^6	B — 18
P1 Q14	(0.07×0.02)÷14	C — 0.0001
P1 Q32	112 flies × 0.1 kg	C — 11.2 kg
P1 Q33	`2^(2m)=4^(14)`	C — 14
P1 Q40	12 cakes equal packets	C — 6
P2 Q7(c)	`(-3)^(x-2)=-243`; `11^(4x-3)` at x=2	x = 7; 1.611×10■
P2 Q5(b)	Journey 11:13→15:08	3 h 55 min
P1 Q3	Match 1:20 pm + 1 h 50	C — 3:10 pm

10. Lesson Summary

Skill	Rule
`a^m × a^n`	add indices
`a^m ÷ a^n`	subtract indices
`(a^m)^n`	multiply indices
`a^(-n)`	`1/a^n`
Decimals add/sub	line up decimal points
Decimals multiply	add the decimal places

Decimals divide	make divisor whole first
Word problems	work carefully; round only at the end

Lesson prepared under `Grade9-Maths/Rules.md` *— plain English, British notation, full workings required.*